

SS2 Part D

Recommendations Report

Account- Activation Prediction

Purpose

This report summarizes results of a viability study performed on a publicly available dataset UCI Bank Marketing with the problem of predicting the term-deposit subscription problem being a close relative of predicting the account activation problem for STADIOEquities. Two models were developed and tested, and this report provides a recommended model, identifies what modifications it requires to fit STADIOEquities data and demonstrates that the results are consistent with those from published research.

1. Which is the best model?

A logistic regression (Model 1) and a random forest (Model 2) were compared on the same pipeline, with the same number of records, and the same hold-out set (20%, 8238 records), where the pipeline was controlled for leakage. The random forest is the recommended model. It also exhibited a better performance than logistic regression in six of the seven metrics, such as the two most relevant for an imbalance ranking problem: ROC-AUC (0.813 versus 0.801) and PR-AUC (0.490 versus 0.461). It also achieved higher precision (0.415 versus 0.366) and F1 (0.503 versus 0.468) at effectively equal recall (0.637 versus 0.649). The practical result is that the random forest found nearly the same number of true subscribers and generated about 200 fewer false positive (832 vs. 1,044) which is the direct result of fewer wasted contact costs per true conversion. Logistic regression should however be considered as an explanatory supplement: the coefficients of logistic regression clearly reflect the direction and strength of the drivers; this is useful for conveying to non-technical stakeholders.

2. How can the selected model be improved?

There are several suggested ways to go about it, most likely in order of value. First, the optimal decision threshold must be used: If the results are above the default 0.5 threshold, the threshold must be optimized by selecting the point on the precision-recall curve that best represents the actual cost of a missed activation in the business. Second, if you're dealing with tabular data, a gradient-boosted tree (like XGBoost or LightGBM) is likely to be a natural next step after a random forest and should be tried. Third, scores should be calibrated (Platt or isotonic scaling) to enable trust in scores for budgeting and prioritization. Fourth, the imbalance could be overcome by incorporating class weighting in addition to resampling techniques like SMOTE. Last, a broader, nested CV hyper-parameter search and more sophisticated feature engineering

(interaction features, recency features) would provide a less biased performance estimate and would probably result in a slight increase in accuracy.

3. Adapting the model to the requested data

The public dataset is only a proxy as several adaptations are necessary for STADIOEquities own data. This should be changed from 'term-deposit subscription' to 'first qualifying deposit within 30 days of registering, based on the registration and first-deposit timestamps. Feature mapping is direct: The campaign-contact fields match the acquisition channel and onboarding nudges; Prior-outcome fields match earlier engagement; Client attributes match the demographic fields requested. Importantly, the leakage discipline employed here, excluding call duration, should be applied to any call duration pre-registered or pre-funded (first-trade date, balance and premium take-up) fields as described in risk R2 of the SS1 RAAIDD log. There are two data challenges that need to be addressed before SS1: Identifying users across app and web sessions and handling of demographic variables in a pseudonymised manner to prevent any unfair discrimination, compliant with POPIA. The conversion has drifted over time (64% to 59%), so a time based split is more appropriate than a random split. The current pipeline will already have the ability to fill in missing values and handle unseen categories and will likely support STADIOEquities' more detailed behavioural signals with minimal modification; it's possible that the signal will be even more powerful than the proxy.

4. Alignment with the related literature

These findings are in good agreement with those found in the Part A literature. The approximate AUC of 0.80-0.81 is similar to Moro, Cortez and Rita (2014) who report AUC of ~0.80 for realistic models without the inclusion of call duration. The performance of the tree ensemble is similar to Elsalamony (2014) who obtained the best results with C5.0 decision tree and neural network, and the high importance of socio-economic indicators among our top features is consistent with that of Moro, Cortez and Rita (2014). Alongside our focus on AUC and precision-recall, the methods we follow are the same ones that are recommended by the literature. The other deliberate difference is we didn't include the duration feature, some studies do and claim greater headline accuracy, but here, deployability was more important than a model that can't be used before a contact with the client is made. The best results obtained by Moro et al. (2011) were from support vector machines, which still are a good option for further research.

Conclusion

The viability study shows that the proposed approach is robust and effective with similar public data: random forest predicts effectively and honestly between subscription/activation, and gives business sense. Having tested and adapted these changes in the above-mentioned manner, STADIOEquities can now confidently use the approach with its own account information.

References

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